

Excitonic coupling of electronic transitions in oligomeric phthalocyanine zinc(II) complexes

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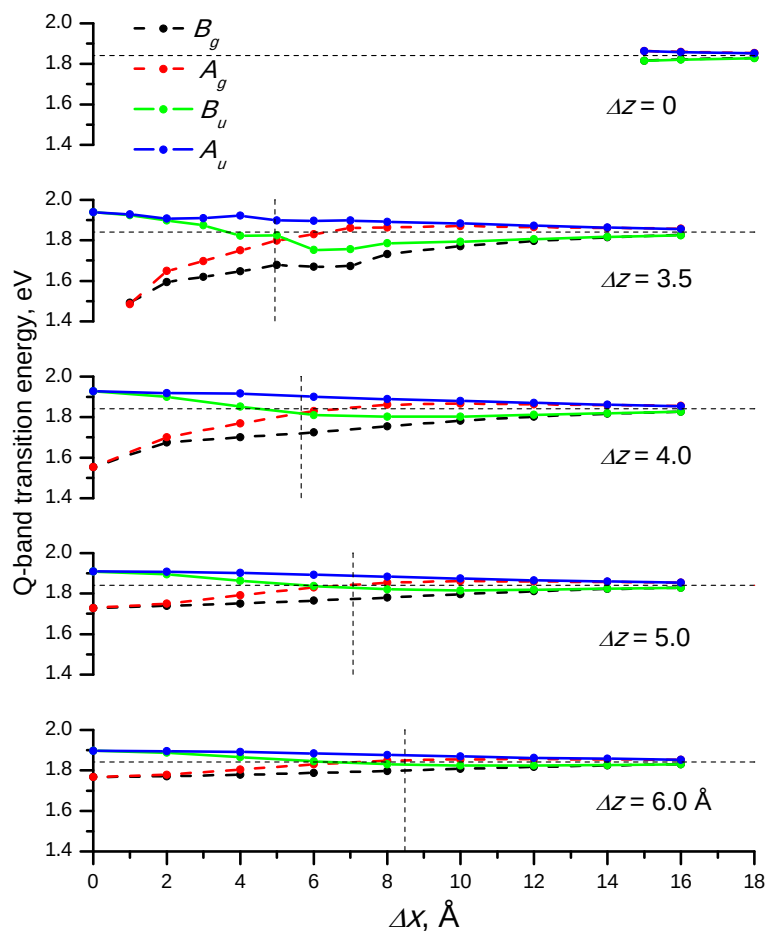


Fig. S1. TD-DFT Q-band transition energies vs. parallel shifts (Δx) of two Pc molecules (arrangement in Fig. 3B of the main text). Forbidden transitions are shown by dashed lines. Horizontal line represents the Q-band energy of ZnPc, vertical line corresponds to zero-splitting angle of $\theta = 54.7^\circ$

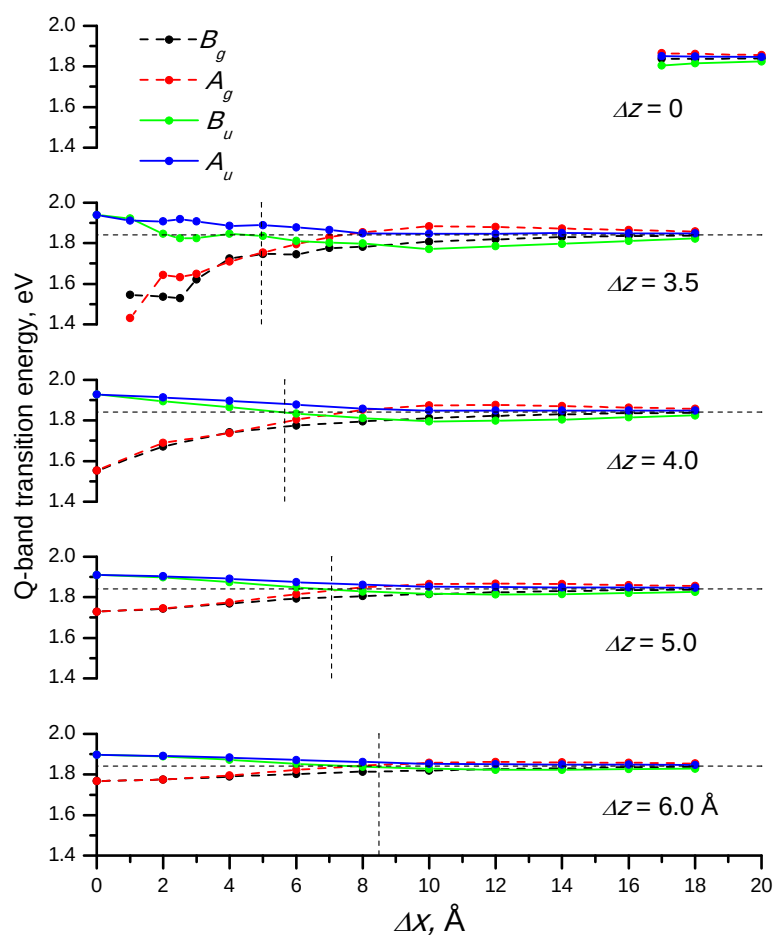


Fig. S2. TD-DFT Q-band transition energies vs. parallel shifts (Δx) of two Pc molecules (arrangement in Fig. 3A of the main text). Forbidden transitions are shown by dashed lines. Horizontal line represents the Q-band energy of ZnPc, vertical line corresponds to zero-splitting angle of $\theta = 54.7^\circ$

Table S1. TD-DFT Q-band transition energies vs. parallel shifts (Δx) of two Pc molecules (arrangement in Fig. 3A of the main text)

Shift, Å		Q-band transition energy, eV				
Δz	Δx	B_g	A_g	$B_u (X)$	$A_u (Y)$	
0	17.0	1.838	1.865	1.804	1.851	
	18.0	1.838	1.861	1.815	1.849	
	20.0	1.839	1.856	1.825	1.847	
3.5	0.0	1.177	1.177	1.940	1.940	
	1.0	1.545	1.432	1.922	1.912	
	2.0	1.537	1.643	1.846	1.907	
	2.5	1.529	1.632	1.824	1.918	
	3.0	1.621	1.649	1.825	1.908	
	4.0	1.725	1.710	1.847	1.885	
	5.0	1.747	1.755	1.835	1.890	
	6.0	1.746	1.796	1.811	1.879	
	7.0	1.777	1.831	1.802	1.866	
	8.0	1.782	1.854	1.799	1.849	
	10.0	1.808	1.883	1.772	1.846	
	12.0	1.819	1.880	1.786	1.846	
	14.0	1.830	1.873	1.797	1.850	
	16.0	1.835	1.865	1.812	1.849	
	18.0	1.837	1.858	1.822	1.848	
4.0	0.0	1.554	1.554	1.928	1.928	
	2.0	1.672	1.690	1.894	1.914	
	4.0	1.742	1.738	1.866	1.896	

	6.0	1.774	1.804	1.833	1.878
	8.0	1.796	1.852	1.812	1.857
	10.0	1.811	1.875	1.796	1.849
	12.0	1.822	1.876	1.798	1.848
	14.0	1.830	1.871	1.805	1.849
	16.0	1.835	1.863	1.815	1.849
	18.0	1.837	1.857	1.824	1.848
5.0	0.0	1.729	1.729	1.910	1.910
	2.0	1.744	1.745	1.898	1.904
	4.0	1.770	1.775	1.874	1.891
	6.0	1.793	1.815	1.848	1.875
	8.0	1.807	1.848	1.829	1.861
	10.0	1.816	1.865	1.817	1.852
	12.0	1.824	1.868	1.814	1.850
	14.0	1.830	1.865	1.816	1.849
	16.0	1.835	1.860	1.821	1.848
	18.0	1.837	1.856	1.827	1.847
6.0	0.0	1.767	1.767	1.896	1.896
	2.0	1.774	1.775	1.889	1.892
	4.0	1.789	1.795	1.872	1.883
	6.0	1.803	1.822	1.853	1.871
	8.0	1.813	1.844	1.838	1.861
	10.0	1.820	1.857	1.828	1.853
	12.0	1.826	1.861	1.823	1.850
	14.0	1.831	1.860	1.823	1.849
	16.0	1.835	1.857	1.826	1.848
	18.0	1.837	1.854	1.829	1.847
8.0	0.0	1.801	1.801	1.876	1.876

* $E(\text{ZnPc}) = 1.842 \text{ eV } (E_U)$

Table S2. TD-DFT Q-band transition energies vs. parallel shifts (Δx) of two Pc molecules (arrangement in Fig. 3B of the main text)

Shift, Å	Q-band transition energy, eV				
	Δx	B_G	A_G	$B_U(X)$	$A_U(Y)$
0	15.0	1.815	1.864	1.816	1.863
	16.0	1.823	1.860	1.821	1.858
	18.0	1.831	1.855	1.828	1.853
3.5	0.0	1.177	1.177	1.940	1.940
	1.0	1.493	1.485	1.924	1.928
	2.0	1.593	1.650	1.898	1.908
	3.0	1.619	1.698	1.874	1.910
	4.0	1.648	1.750	1.822	1.922
	5.0	1.678	1.799	1.824	1.898
	6.0	1.669	1.830	1.752	1.896
	7.0	1.673	1.862	1.756	1.899
	8.0	1.733	1.864	1.785	1.892
	10.0	1.771	1.870	1.794	1.883
	12.0	1.797	1.866	1.806	1.873
	14.0	1.816	1.862	1.818	1.863
	16.0	1.826	1.857	1.825	1.856
4.0	0.0	1.554	1.554	1.928	1.928
	2.0	1.675	1.701	1.900	1.919
	4.0	1.701	1.770	1.852	1.917
	6.0	1.725	1.831	1.809	1.901
	8.0	1.754	1.861	1.803	1.889
	10.0	1.782	1.867	1.803	1.880
	12.0	1.803	1.864	1.811	1.870
	14.0	1.818	1.860	1.820	1.862
	16.0	1.827	1.856	1.826	1.855
5.0	0.0	1.729	1.729	1.910	1.910
	2.0	1.739	1.751	1.895	1.908
	4.0	1.751	1.792	1.864	1.903
	6.0	1.766	1.831	1.837	1.893
	8.0	1.781	1.854	1.821	1.883
	10.0	1.797	1.861	1.816	1.874

	12.0	1.811	1.860	1.819	1.866
	14.0	1.822	1.858	1.824	1.859
	16.0	1.829	1.855	1.828	1.854
6.0	0.0	1.767	1.767	1.896	1.896
	2.0	1.771	1.778	1.887	1.895
	4.0	1.778	1.804	1.866	1.891
	6.0	1.787	1.831	1.845	1.884
	8.0	1.797	1.848	1.831	1.876
	10.0	1.808	1.855	1.825	1.869
	12.0	1.817	1.856	1.825	1.862
	14.0	1.825	1.855	1.827	1.857
	16.0	1.830	1.853	1.830	1.853

* $E(\text{ZnPc}) = 1.842 \text{ eV } (E_U)$

Table S3. Major contributions of the elementary MO-MO transitions to the Q-band transitions of face-to-face ZnPc dimers. Allowed transitions are shown in bold

$\Delta z, \text{\AA}$	Transition		Occupied MO		Virtual MO		Contribution
	Symmetry	Energy, eV	Symmetry	Energy, eV	Symmetry	Energy, eV	
3.5	E_G	1.177	a_{2g}	-6.794	e_g	-1.254	4.1%
			a_{1u}	-6.060	e_u	-1.908	94.4%
	E_U	1.940	a_{2g}	-6.794	e_u	-1.908	38.1%
			a_{1u}	-6.060	e_g	-1.254	53.0%
4.0	E_G	1.554	a_{2g}	-6.602	e_g	-1.436	16.6%
			a_{1u}	-6.265	e_u	-1.745	80.2%
	E_U	1.928	a_{2g}	-6.602	e_u	-1.745	44.3%
			a_{1u}	-6.265	e_g	-1.436	47.0%
5.0	E_G	1.729	a_{2g}	-6.489	e_g	-1.579	41.1%
			a_{1u}	-6.422	e_u	-1.643	53.9%
	E_U	1.910	a_{2g}	-6.489	e_u	-1.643	45.8%
			a_{1u}	-6.422	e_g	-1.579	45.9%
6.0	E_G	1.767	a_{2g}	-6.481	e_g	-1.622	46.2%
			a_{1u}	-6.468	e_u	-1.635	48.3%
	E_U	1.896	a_{2g}	-6.481	e_u	-1.635	46.0%
			a_{1u}	-6.468	e_g	-1.622	46.0%

Table S4A. MO energies of ZnPc dimers at $\Delta z = 3.5 \text{ \AA}$, eV (MO description is shown in Fig. 5, monomer arrangement in Fig. 3A of the main text)

$\Delta x, \text{\AA}$	Occupied MO			Virtual MO		
	b_g	a_u	a_u	b_u	a_g	b_g
0	-6.794	-6.060	-1.908	-1.908	-1.254	-1.254
1	-6.617	-6.243	-1.804	-1.675	-1.512	-1.384
2	-6.340	-6.539	-1.672	-1.419	-1.785	-1.563
3	-6.287	-6.634	-1.679	-1.533	-1.684	-1.598
4	-6.406	-6.561	-1.721	-1.679	-1.568	-1.581
5	-6.466	-6.548	-1.713	-1.641	-1.649	-1.620
6	-6.440	-6.605	-1.709	-1.598	-1.735	-1.642
7	-6.467	-6.603	-1.725	-1.655	-1.714	-1.641
8	-6.562	-6.514	-1.687	-1.730	-1.686	-1.691
10	-6.619	-6.489	-1.609	-1.717	-1.727	-1.794
12	-6.526	-6.584	-1.722	-1.730	-1.716	-1.694
14	-6.550	-6.581	-1.743	-1.720	-1.724	-1.694
16	-6.568	-6.572	-1.729	-1.722	-1.725	-1.720

Table S4B. MO energies of ZnPc dimers at $\Delta z = 3.5 \text{ \AA}$, eV (MO description is shown in Fig. 5, monomer arrangement in Fig. 3B of the main text)

$\Delta x, \text{\AA}$	Occupied MO			Virtual MO		
	b_g	a_u	a_u	b_u	a_g	b_g
0	-6.794	-6.060	-1.908	-1.908	-1.254	-1.254
1	-6.621	-6.234	-1.736	-1.738	-1.448	-1.453
2	-6.400	-6.481	-1.546	-1.486	-1.733	-1.674
3	-6.448	-6.486	-1.570	-1.497	-1.754	-1.683

4	-6.548	-6.394	-1.594	-1.672	-1.605	-1.681
5	-6.467	-6.506	-1.630	-1.730	-1.579	-1.653
6	-6.324	-6.665	-1.756	-1.623	-1.689	-1.563
7	-6.370	-6.663	-1.802	-1.589	-1.772	-1.540
8	-6.486	-6.554	-1.755	-1.648	-1.717	-1.610
10	-6.531	-6.558	-1.718	-1.707	-1.696	-1.690
12	-6.544	-6.578	-1.727	-1.700	-1.731	-1.707
14	-6.567	-6.572	-1.725	-1.721	-1.727	-1.722
16	-6.569	-6.569	-1.722	-1.722	-1.723	-1.722

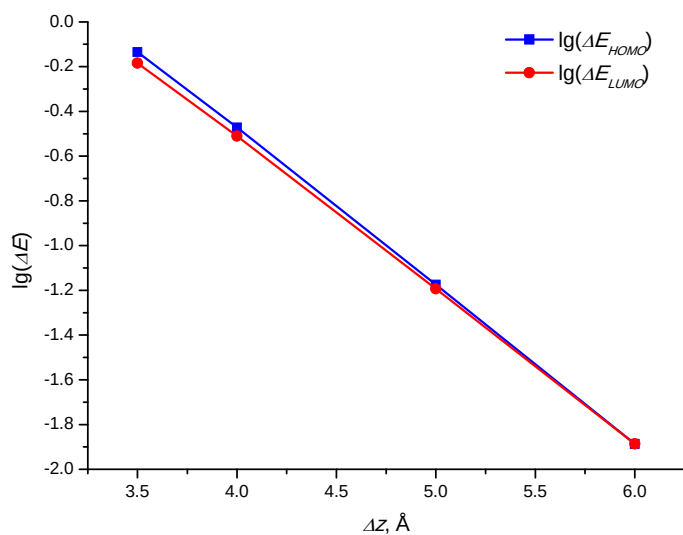


Fig. S3. Distance dependence of HOMO and LUMO splitting for face-to-face ZnPc dimers ($\Delta x = 0$)

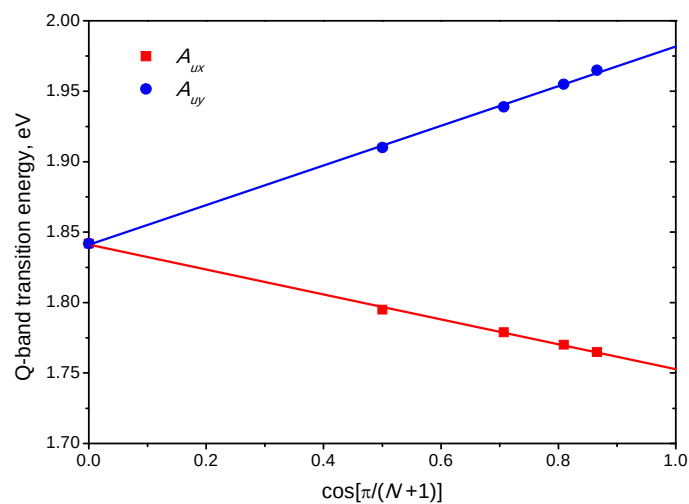


Fig. S4. Size dependence of the main TD-DFT Q-band transitions of slipped-stacked ZnPc oligomers of size N (resembling β -ZnPc stacks)

Table S5. Size dependence of TD-DFT energies of the main Q-band transitions of slipped-stacked ZnPc oligomers

Shifts, Å	Number of monomers	Q-band transition energies, eV, and oscillator strengths			
Along Zn – meso-N (Fig. 3B)		$E [A_u (X)]$	$E [A_u (Y)]$	$f (A_{ux})$	$f (A_{uy})$
$\Delta x = 3.632$	1	1.842	1.842	0.472	0.472

$\Delta y = 0.185$	2	1.795	1.910	0.685	0.668
$\Delta z = 3.214$	3	1.779	1.939	0.957	0.826
(β -ZnPc stack)	4	1.770	1.955	1.231	0.973
	5	1.765	1.965	1.428	1.114
	∞	1.752	1.982	1.547	1 (ratio)
$\Delta x = 1.75$	1	1.842	1.842	0.472	0.472
$\Delta y = 0$	2	1.896	1.900	0.683	0.606
$\Delta z = 3.4$	3	1.928	1.930	0.859	0.706
(approx. α -ZnPc stack)	4	1.943	1.943	1.036	0.790
	∞	1.963	1.965	1.775	1 (ratio)
$\Delta x = 2.0$	1	1.842	1.842	0.472	0.472
$\Delta y = 0$	2	1.896	1.900	0.683	0.606
$\Delta z = 3.4$	3	1.920	1.923	0.874	0.705
(approx. α -ZnPc stack)	4	1.932	1.935	1.061	0.798
	∞	1.953	1.956	1.819	1 (ratio)
Along Zn–N bond (Fig. 3A)		$E [B_u (X)]$	$E [A_u (Y)]$	$f (B_u)$	$f (A_u)$
$\Delta x = 1.75$	1	1.842	1.842	0.472	0.472
$\Delta y = 0$	2	1.820	1.872	0.524	0.564
$\Delta z = 3.3$	3	1.824	1.889	0.623	0.662
(approx. α -ZnPc stack)	4	1.825	1.897	0.723	0.756
	∞	1.816	1.909	1 (ratio)	1.115
$\Delta x = 2.0$	1	1.842	1.842	0.472	0.472
$\Delta y = 0$	2	1.791	1.891	0.521	0.598
$\Delta z = 3.3$	3	1.782	1.912	0.618	0.715
(approx. α -ZnPc stack)	4	1.778	1.924	0.730	0.828
	∞	1.758	1.942	1 (ratio)	1.357
$\Delta x = 1.75$	1	1.842	1.842	0.472	0.472
$\Delta y = 0$	2	1.846	1.889	0.568	0.599
$\Delta z = 3.4$	3	1.854	1.910	0.682	0.712
(approx. α -ZnPc stack)	4	1.859	1.921	0.795	0.821
	∞	1.860	1.939	1 (ratio)	1.070
$\Delta x = 2.0$	1	1.842	1.842	0.472	0.472
$\Delta y = 0$	2	1.823	1.901	0.563	0.623
$\Delta z = 3.4$	3	1.821	1.926	0.679	0.754
(approx. α -ZnPc stack)	4	1.820	1.939	0.805	0.881
	∞	1.812	1.961	1 (ratio)	1.217

XYZ geometries optimized at the r2SCAN-3c level of DFT, Å

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ZnPc

C	0.701663	6.554204	0.000000
C	-0.701663	6.554204	0.000000
C	1.422571	5.364926	0.000000
C	0.703967	4.174565	0.000000
C	-0.703967	4.174565	0.000000
C	-1.422571	5.364926	0.000000
C	1.120422	2.780020	0.000000
C	-1.120422	2.780020	0.000000
N	0.000000	1.989101	0.000000
N	-2.390568	2.390568	0.000000
N	2.390568	2.390568	0.000000
H	1.231071	7.502550	0.000000
H	-1.231071	7.502550	0.000000
H	2.508027	5.356475	0.000000
H	-2.508027	5.356475	0.000000
C	6.554204	-0.701663	0.000000
C	6.554204	0.701663	0.000000
C	5.364926	-1.422571	0.000000

C	4.174565	-0.703967	0.000000
C	4.174565	0.703967	0.000000
C	5.364926	1.422571	0.000000
C	2.780020	-1.120422	0.000000
C	2.780020	1.120422	0.000000
N	1.989101	0.000000	0.000000
N	2.390568	-2.390568	0.000000
H	7.502550	-1.231071	0.000000
H	7.502550	1.231071	0.000000
H	5.356475	-2.508027	0.000000
H	5.356475	2.508027	0.000000
C	-6.554204	0.701663	0.000000
C	-6.554204	-0.701663	0.000000
C	-5.364926	1.422571	0.000000
C	-4.174565	0.703967	0.000000
C	-4.174565	-0.703967	0.000000
C	-5.364926	-1.422571	0.000000
C	-2.780020	1.120422	0.000000
C	-2.780020	-1.120422	0.000000
N	-1.989101	0.000000	0.000000
N	-2.390568	-2.390568	0.000000
H	-7.502550	1.231071	0.000000
H	-7.502550	-1.231071	0.000000
H	-5.356475	2.508027	0.000000
H	-5.356475	-2.508027	0.000000
C	-0.701663	-6.554204	0.000000
C	0.701663	-6.554204	0.000000
C	-1.422571	-5.364926	0.000000
C	-0.703967	-4.174565	0.000000
C	0.703967	-4.174565	0.000000
C	1.422571	-5.364926	0.000000
C	-1.120422	-2.780020	0.000000
C	1.120422	-2.780020	0.000000
N	-0.000000	-1.989101	0.000000
H	-1.231071	-7.502550	0.000000
H	1.231071	-7.502550	0.000000
H	-2.508027	-5.356475	0.000000
H	2.508027	-5.356475	0.000000
Zn	0.000000	0.000000	0.000000

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Compound **1**

C	0.032844	1.293259	3.255751
C	0.032844	-1.293259	3.255751
H	0.032910	-2.387650	3.225592
H	0.032910	2.387650	3.225592
C	1.261492	0.706647	2.602144
C	2.316244	1.429237	2.068434
C	1.261492	-0.706647	2.602144
H	2.322996	2.515722	2.064498
C	2.316244	-1.429237	2.068434
H	2.322996	-2.515722	2.064498
C	-1.176886	0.706455	2.571257
C	-1.176886	-0.706455	2.571257
C	-2.222960	1.429234	2.023244
C	-2.222960	-1.429234	2.023244

H	-2.231818	2.515768	2.023710
H	-2.231818	-2.515768	2.023710
C	-3.275706	0.703533	1.469838
C	-3.275706	-0.703533	1.469838
C	-4.511329	1.119495	0.826224
C	-4.511329	-1.119495	0.826224
N	-4.856526	2.390259	0.647421
N	-5.214860	0.000000	0.460990
N	-4.856526	-2.390259	0.647421
C	-5.985708	2.779972	0.065649
Zn	-6.982533	0.000000	-0.445487
C	-5.985708	-2.779972	0.065649
C	-6.355226	4.174640	-0.125806
N	-6.982989	1.989976	-0.445560
N	-8.752521	0.000000	-1.354016
N	-6.982989	-1.989976	-0.445560
C	-6.355226	-4.174640	-0.125806
C	-5.714869	5.365018	0.200409
C	-7.608323	4.175087	-0.767982
C	-7.979554	2.780647	-0.957303
C	-9.456343	1.120515	-1.714255
C	-9.456343	-1.120515	-1.714255
C	-7.979554	-2.780647	-0.957303
C	-7.608323	-4.175087	-0.767982
C	-5.714869	-5.365018	0.200409
C	-6.355392	6.554629	-0.129271
H	-4.748426	5.356380	0.694648
C	-8.246937	5.365829	-1.096284
N	-9.109798	2.390926	-1.536574
C	-10.697446	0.704007	-2.349670
C	-10.697446	-0.704007	-2.349670
N	-9.109798	-2.390926	-1.536574
C	-8.246937	-5.365829	-1.096284
C	-6.355392	-6.554629	-0.129271
H	-4.748426	-5.356380	0.694648
C	-7.604577	6.555024	-0.768614
H	-5.883248	7.502827	0.110896
H	-9.213140	5.357964	-1.590994
C	-11.757658	1.422404	-2.891399
C	-11.757658	-1.422404	-2.891399
C	-7.604577	-6.555024	-0.768614
H	-9.213140	-5.357964	-1.590994
H	-5.883248	-7.502827	0.110896
H	-8.075213	7.503446	-1.010772
C	-12.816710	0.701631	-3.432620
H	-11.750276	2.507884	-2.887344
C	-12.816710	-0.701631	-3.432620
H	-11.750276	-2.507884	-2.887344
H	-8.075213	-7.503446	-1.010772
H	-13.661115	1.231131	-3.864161
H	-13.661115	-1.231131	-3.864161
C	3.376656	-0.703747	1.528442
C	3.376656	0.703747	1.528442
C	4.618790	-1.119718	0.897231
C	4.618790	1.119718	0.897231
N	4.965979	-2.390526	0.722396

N	5.325329	0.000000	0.538641
N	4.965979	2.390526	0.722396
C	6.099947	-2.780356	0.150296
Zn	7.100112	0.000000	-0.355230
C	6.099947	2.780356	0.150296
C	6.471609	-4.174936	-0.037275
N	7.101118	-1.989980	-0.353093
N	8.878220	0.000000	-1.246810
N	7.101118	1.989980	-0.353093
C	6.471609	4.174936	-0.037275
C	5.828916	-5.365431	0.283838
C	7.729980	-4.175093	-0.669014
C	8.102142	-2.780587	-0.856011
C	9.585038	-1.120565	-1.601217
C	9.585038	1.120565	-1.601217
C	8.102142	2.780587	-0.856011
C	7.729980	4.175093	-0.669014
C	5.828916	5.365431	0.283838
C	6.472719	-6.554865	-0.040127
H	4.858124	-5.357231	0.769475
C	8.371836	-5.365621	-0.991625
N	9.237198	-2.390821	-1.425989
C	10.831376	-0.704040	-2.226580
C	10.831376	0.704040	-2.226580
N	9.237198	2.390821	-1.425989
C	8.371836	5.365621	-0.991625
C	6.472719	6.554865	-0.040127
H	4.858124	5.357231	0.769475
C	7.727199	-6.554961	-0.668929
H	5.998800	-7.503158	0.196056
H	9.342039	-5.357570	-1.478423
C	11.895742	-1.422561	-2.759760
C	11.895742	1.422561	-2.759760
C	7.727199	6.554961	-0.668929
H	9.342039	5.357570	-1.478423
H	5.998800	7.503158	0.196056
H	8.200111	-7.503267	-0.907004
C	12.959096	-0.701692	-3.292554
H	11.888281	-2.508028	-2.755891
C	12.959096	0.701692	-3.292554
H	11.888281	2.508028	-2.755891
H	8.200111	7.503267	-0.907004
H	13.806851	-1.231185	-3.717460
H	13.806851	1.231185	-3.717460
C	0.035209	-0.778597	4.733061
C	-1.240053	-1.137349	5.514324
C	0.035209	0.778597	4.733061
H	0.927158	-1.176667	5.227701
O	-2.094550	0.000000	5.427058
H	-0.996340	-1.353478	6.568635
H	-1.781594	-1.992215	5.094259
C	-1.240053	1.137349	5.514324
H	0.927158	1.176667	5.227701
H	-1.781594	1.992215	5.094259
H	-0.996340	1.353478	6.568635

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Compound 2

C	-1.274825	0.704246	0.000000
C	-1.274825	-0.704246	0.000000
C	0.000000	1.539445	0.000000
C	0.000000	2.430155	1.266875
C	0.000000	2.430155	-1.266875
C	0.000000	-1.539445	0.000000
C	0.000000	-2.430155	-1.266875
C	0.000000	-2.430155	1.266875
C	1.274825	-0.704246	0.000000
C	1.274825	0.704246	0.000000
C	-8.559367	-6.554527	0.000000
C	-9.280464	-5.365374	0.000000
C	-8.562211	-4.174734	0.000000
C	-8.978893	-2.780539	0.000000
C	-8.559367	6.554527	0.000000
C	-9.280464	5.365374	0.000000
C	-8.562211	4.174734	0.000000
C	-8.978893	2.780539	0.000000
C	-7.155996	-6.554441	0.000000
C	-6.435431	-5.365043	0.000000
C	-7.154186	-4.174638	0.000000
C	-6.737976	-2.780487	0.000000
C	-7.155996	6.554441	0.000000
C	-6.435431	5.365043	0.000000
C	-7.154186	4.174638	0.000000
C	-6.737976	2.780487	0.000000
C	-14.411901	-0.701738	0.000000
C	-13.222795	-1.422611	0.000000
C	-12.032126	-0.704108	0.000000
C	-10.638236	-1.120416	0.000000
C	-14.411901	0.701738	0.000000
C	-13.222795	1.422611	0.000000
C	-12.032126	0.704108	0.000000
C	-10.638236	1.120416	0.000000
C	-2.493769	-1.400462	0.000000
C	-3.684024	-0.702038	0.000000
C	-5.077541	-1.121943	0.000000
C	-2.493769	1.400462	0.000000
C	-3.684024	0.702038	0.000000
C	-5.077541	1.121943	0.000000
N	-7.857207	-1.989394	0.000000
N	-7.857207	1.989394	0.000000
N	-9.847170	0.000000	0.000000
N	-5.868105	0.000000	0.000000
N	-10.248346	-2.391044	0.000000
N	-10.248346	2.391044	0.000000
N	-5.466545	-2.391439	0.000000
N	-5.466545	2.391439	0.000000
Zn	-7.858269	0.000000	0.000000
C	7.155996	-6.554441	0.000000
C	6.435431	-5.365043	0.000000
C	7.154186	-4.174638	0.000000
C	6.737976	-2.780487	0.000000
C	7.155996	6.554441	0.000000

C	6.435431	5.365043	0.000000
C	7.154186	4.174638	0.000000
C	6.737976	2.780487	0.000000
C	8.559367	-6.554527	0.000000
C	9.280464	-5.365374	0.000000
C	8.562211	-4.174734	0.000000
C	8.978893	-2.780539	0.000000
C	8.559367	6.554527	0.000000
C	9.280464	5.365374	0.000000
C	8.562211	4.174734	0.000000
C	8.978893	2.780539	0.000000
C	2.493769	-1.400462	0.000000
C	3.684024	-0.702038	0.000000
C	5.077541	-1.121943	0.000000
C	2.493769	1.400462	0.000000
C	3.684024	0.702038	0.000000
C	5.077541	1.121943	0.000000
C	14.411901	-0.701738	0.000000
C	13.222795	-1.422611	0.000000
C	12.032126	-0.704108	0.000000
C	10.638236	-1.120416	0.000000
C	14.411901	0.701738	0.000000
C	13.222795	1.422611	0.000000
C	12.032126	0.704108	0.000000
C	10.638236	1.120416	0.000000
N	7.857207	-1.989394	0.000000
N	7.857207	1.989394	0.000000
N	5.868105	0.000000	0.000000
N	9.847170	0.000000	0.000000
N	5.466545	-2.391439	0.000000
N	5.466545	2.391439	0.000000
N	10.248346	-2.391044	0.000000
N	10.248346	2.391044	0.000000
Zn	7.858269	0.000000	0.000000
H	0.882943	3.074398	1.296549
H	-0.882943	3.074398	1.296549
H	0.000000	1.807752	2.167250
H	0.882943	3.074398	-1.296549
H	0.000000	1.807752	-2.167250
H	-0.882943	3.074398	-1.296549
H	0.882943	-3.074398	-1.296549
H	-0.882943	-3.074398	-1.296549
H	0.000000	-1.807752	-2.167250
H	0.882943	-3.074398	1.296549
H	0.000000	-1.807752	2.167250
H	-0.882943	-3.074398	1.296549
H	-9.088652	-7.502952	0.000000
H	-10.365919	-5.357110	0.000000
H	-9.088652	7.502952	0.000000
H	-10.365919	5.357110	0.000000
H	-6.626545	-7.502778	0.000000
H	-5.349961	-5.356397	0.000000
H	-6.626545	7.502778	0.000000
H	-5.349961	5.356397	0.000000
H	-15.360307	-1.231067	0.000000
H	-13.214403	-2.508074	0.000000

H	-15.360307	1.231067	0.000000
H	-13.214403	2.508074	0.000000
H	-2.521590	-2.484957	0.000000
H	-2.521590	2.484957	0.000000
H	6.626545	-7.502778	0.000000
H	5.349961	-5.356397	0.000000
H	6.626545	7.502778	0.000000
H	5.349961	5.356397	0.000000
H	9.088652	-7.502952	0.000000
H	10.365919	-5.357110	0.000000
H	9.088652	7.502952	0.000000
H	10.365919	5.357110	0.000000
H	2.521590	-2.484957	0.000000
H	2.521590	2.484957	0.000000
H	15.360307	-1.231067	0.000000
H	13.214403	-2.508074	0.000000
H	15.360307	1.231067	0.000000
H	13.214403	2.508074	0.000000

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Compound **3**

C	4.568084	6.546983	-0.480033
C	5.937489	6.550230	-0.173666
C	3.874312	5.355961	-0.665606
C	4.584870	4.167795	-0.536170
C	5.958904	4.171117	-0.229528
C	6.650554	5.363140	-0.044437
C	4.188621	2.772481	-0.660137
C	6.375068	2.777735	-0.167728
N	5.286965	1.984946	-0.431197
N	7.615923	2.390324	0.103196
N	2.951718	2.385795	-0.945800
H	4.044193	7.493673	-0.574179
H	6.446824	7.499567	-0.036197
H	2.815394	5.345656	-0.905413
H	7.709921	5.357685	0.192123
C	-1.124209	-0.699112	-1.747139
C	-1.124209	0.699112	-1.747139
C	0.007360	-1.437167	-1.387736
C	1.199487	-0.706906	-1.340176
C	1.199487	0.706906	-1.340176
C	0.007360	1.437167	-1.387736
C	2.564525	-1.115977	-1.053167
C	2.564525	1.115977	-1.053167
N	3.347255	0.000000	-0.914279
N	2.951718	-2.385795	-0.945800
H	-2.072126	-1.219467	-1.867814
H	-2.072126	1.219467	-1.867814
C	11.685884	0.701508	0.982177
C	11.685884	-0.701508	0.982177
C	10.524007	1.422448	0.728585
C	9.361210	0.703906	0.473542
C	9.361210	-0.703906	0.473542
C	10.524007	-1.422448	0.728585
C	7.999037	1.119768	0.176023
C	7.999037	-1.119768	0.176023

N	7.226733	0.000000	0.003489
N	7.615923	-2.390324	0.103196
H	12.612360	1.230916	1.184510
H	12.612360	-1.230916	1.184510
H	10.515783	2.507922	0.727440
H	10.515783	-2.507922	0.727440
C	5.937489	-6.550230	-0.173666
C	4.568084	-6.546983	-0.480033
C	6.650554	-5.363140	-0.044437
C	5.958904	-4.171117	-0.229528
C	4.584870	-4.167795	-0.536170
C	3.874312	-5.355961	-0.665606
C	6.375068	-2.777735	-0.167728
C	4.188621	-2.772481	-0.660137
N	5.286965	-1.984946	-0.431197
H	6.446824	-7.499567	-0.036197
H	4.044193	-7.493673	-0.574179
H	7.709921	-5.357685	0.192123
H	2.815394	-5.345656	-0.905413
Zn	5.286575	0.000000	-0.460123
C	-0.132511	2.811888	-0.784795
H	0.569314	3.521396	-1.231920
H	-1.151731	3.166452	-0.964905
C	-0.132511	-2.811888	-0.784795
H	0.569314	-3.521396	-1.231920
H	-1.151731	-3.166452	-0.964905
C	-4.568084	6.546983	0.480033
C	-5.937489	6.550230	0.173666
C	-3.874312	5.355961	0.665606
C	-4.584870	4.167795	0.536170
C	-5.958904	4.171117	0.229528
C	-6.650554	5.363140	0.044437
C	-4.188621	2.772481	0.660137
C	-6.375068	2.777735	0.167728
N	-5.286965	1.984946	0.431197
N	-7.615923	2.390324	-0.103196
N	-2.951718	2.385795	0.945800
H	-4.044193	7.493673	0.574179
H	-6.446824	7.499567	0.036197
H	-2.815394	5.345656	0.905413
H	-7.709921	5.357685	-0.192123
C	1.124209	-0.699112	1.747139
C	1.124209	0.699112	1.747139
C	-0.007360	-1.437167	1.387736
C	-1.199487	-0.706906	1.340176
C	-1.199487	0.706906	1.340176
C	-0.007360	1.437167	1.387736
C	-2.564525	-1.115977	1.053167
C	-2.564525	1.115977	1.053167
N	-3.347255	0.000000	0.914279
N	-2.951718	-2.385795	0.945800
H	2.072126	-1.219467	1.867814
H	2.072126	1.219467	1.867814
C	-11.685884	0.701508	-0.982177
C	-11.685884	-0.701508	-0.982177
C	-10.524007	1.422448	-0.728585

C	-9.361210	0.703906	-0.473542
C	-9.361210	-0.703906	-0.473542
C	-10.524007	-1.422448	-0.728585
C	-7.999037	1.119768	-0.176023
C	-7.999037	-1.119768	-0.176023
N	-7.226733	0.000000	-0.003489
N	-7.615923	-2.390324	-0.103196
H	-12.612360	1.230916	-1.184510
H	-12.612360	-1.230916	-1.184510
H	-10.515783	2.507922	-0.727440
H	-10.515783	-2.507922	-0.727440
C	-5.937489	-6.550230	0.173666
C	-4.568084	-6.546983	0.480033
C	-6.650554	-5.363140	0.044437
C	-5.958904	-4.171117	0.229528
C	-4.584870	-4.167795	0.536170
C	-3.874312	-5.355961	0.665606
C	-6.375068	-2.777735	0.167728
C	-4.188621	-2.772481	0.660137
N	-5.286965	-1.984946	0.431197
H	-6.446824	-7.499567	0.036197
H	-4.044193	-7.493673	0.574179
H	-7.709921	-5.357685	-0.192123
H	-2.815394	-5.345656	0.905413
Zn	-5.286575	0.000000	0.460123
C	0.132511	2.811888	0.784795
H	-0.569314	3.521396	1.231920
H	1.151731	3.166452	0.964905
C	0.132511	-2.811888	0.784795
H	-0.569314	-3.521396	1.231920
H	1.151731	-3.166452	0.964905

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Compound 6

C	4.799976	-6.365855	1.346691
C	6.173650	-6.199617	1.117098
C	3.935208	-5.275401	1.365364
C	4.479020	-4.015833	1.143325
C	5.857376	-3.848968	0.914220
C	6.720336	-4.938210	0.899822
C	3.892144	-2.681371	1.102435
C	6.072417	-2.418113	0.730857
N	4.879028	-1.769005	0.855149
N	7.269251	-1.886920	0.486615
N	2.598321	-2.446899	1.303677
H	4.408170	-7.364811	1.515737
H	6.819711	-7.072849	1.112722
H	2.871681	-5.395273	1.548027
H	7.783224	-4.800341	0.726818
C	-1.800357	0.180813	2.141665
C	-1.643122	-1.220906	2.150310
C	-0.717108	1.019635	1.864308
C	0.509936	0.428341	1.609767
C	0.670910	-0.969912	1.645675
C	-0.407010	-1.806453	1.912982
C	1.797757	0.985773	1.226393

C	2.066130	-1.227589	1.318805
N	2.686699	-0.036291	1.062351
N	2.005185	2.291049	1.074480
H	-2.513073	-1.840919	2.348070
H	-0.835382	2.097975	1.800867
H	-0.287365	-2.885118	1.915596
C	11.146884	0.289970	-0.149833
C	10.956755	1.671556	-0.300176
C	10.070128	-0.567199	0.056746
C	8.798867	-0.007225	0.102091
C	8.608258	1.378582	-0.048359
C	9.684971	2.234222	-0.247968
C	7.475153	-0.585826	0.303869
C	7.173344	1.612216	0.066736
N	6.543612	0.414961	0.262454
N	6.627952	2.824013	0.017607
H	12.154209	-0.114089	-0.193439
H	11.820554	2.311056	-0.457546
H	10.208611	-1.636816	0.180539
H	9.529889	3.303042	-0.357980
C	4.494163	6.767568	0.275734
C	3.136249	6.612037	0.591306
C	5.330582	5.665188	0.128297
C	4.771991	4.403604	0.295736
C	3.410458	4.247506	0.614497
C	2.577946	5.349794	0.769454
C	5.335650	3.060241	0.226948
C	3.176902	2.813276	0.730427
N	4.343598	2.151326	0.463509
H	4.896820	7.768075	0.146660
H	2.513122	7.495180	0.699652
H	6.383689	5.777254	-0.110286
H	1.529034	5.220867	1.018579
Zn	4.522123	0.143852	0.195515
C	-3.119283	0.745108	2.429904
C	-4.711703	1.930431	3.411686
C	-5.257597	1.144588	2.440428
H	-5.153131	2.631209	4.103915
H	-6.290469	1.071143	2.131202
N	-4.264137	0.409452	1.832514
N	-3.357158	1.671636	3.401341
C	-2.388709	2.255935	4.322086
H	-2.851589	2.363876	5.306292
H	-1.521205	1.598246	4.403012
H	-2.057783	3.238366	3.968934
C	1.800357	-0.180813	-2.141665
C	1.643122	1.220906	-2.150310
C	0.717108	-1.019635	-1.864308
C	0.407010	1.806453	-1.912982
H	2.513073	1.840919	-2.348070
C	-0.509936	-0.428341	-1.609767
H	0.835382	-2.097975	-1.800867
C	-0.670910	0.969912	-1.645675
H	0.287365	2.885118	-1.915596
C	-1.797757	-0.985773	-1.226393
C	-2.066130	1.227589	-1.318805

N	-2.686699	0.036291	-1.062351
N	-2.005185	-2.291049	-1.074480
N	-2.598321	2.446899	-1.303677
Zn	-4.522123	-0.143852	-0.195515
C	-3.176902	-2.813276	-0.730427
C	-3.892144	2.681371	-1.102435
N	-4.879028	1.769005	-0.855149
N	-6.543612	-0.414961	-0.262454
N	-4.343598	-2.151326	-0.463509
C	-3.410458	-4.247506	-0.614497
C	-4.479020	4.015833	-1.143325
C	-6.072417	2.418113	-0.730857
C	-7.475153	0.585826	-0.303869
C	-7.173344	-1.612216	-0.066736
C	-5.335650	-3.060241	-0.226948
C	-4.771991	-4.403604	-0.295736
C	-2.577946	-5.349794	-0.769454
C	-3.935208	5.275401	-1.365364
C	-5.857376	3.848968	-0.914220
N	-7.269251	1.886920	-0.486615
C	-8.798867	0.007225	-0.102091
C	-8.608258	-1.378582	0.048359
N	-6.627952	-2.824013	-0.017607
C	-5.330582	-5.665188	-0.128297
C	-3.136249	-6.612037	-0.591306
H	-1.529034	-5.220867	-1.018579
C	-4.799976	6.365855	-1.346691
H	-2.871681	5.395273	-1.548027
C	-6.720336	4.938210	-0.899822
C	-10.070128	0.567199	-0.056746
C	-9.684971	-2.234222	0.247968
C	-4.494163	-6.767568	-0.275734
H	-6.383689	-5.777254	0.110286
H	-2.513122	-7.495180	-0.699652
C	-6.173650	6.199617	-1.117098
H	-4.408170	7.364811	-1.515737
H	-7.783224	4.800341	-0.726818
C	-11.146884	-0.289970	0.149833
H	-10.208611	1.636816	-0.180539
C	-10.956755	-1.671556	0.300176
H	-9.529889	-3.303042	0.357980
H	-4.896820	-7.768075	-0.146660
H	-6.819711	7.072849	-1.112722
H	-12.154209	0.114089	0.193439
H	-11.820554	-2.311056	0.457546
C	3.119283	-0.745108	-2.429904
N	4.264137	-0.409452	-1.832514
N	3.357158	-1.671636	-3.401341
C	5.257597	-1.144588	-2.440428
C	4.711703	-1.930431	-3.411686
C	2.388709	-2.255935	-4.322086
H	6.290469	-1.071143	-2.131202
H	5.153131	-2.631209	-4.103915
H	2.057783	-3.238366	-3.968934
H	1.521205	-1.598246	-4.403012
H	2.851589	-2.363876	-5.306292